



FD24-053-D-VoiceGrocery

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1 Introduction

In today's modern digital world, where everything is altered, and almost every product is just a click away, some illiterate people still find grocery shopping difficult. Because these grocery apps rely heavily on text-based guidance. Which increases the difficulty for such people when it comes to shopping for food online. Voice Grocery, solves this major issue by offering simple voice-based navigation and virtual assistant where user can control the whole process through simple voice commands, and the user can search for the specific process, add it to cart then check out, moreover user can also give audio feedback about the product he/she have received. This strategy bridges gap in engagement of users in daily routine tasks such as food and promotes independence and autonomy among the users.

1.1 Problem Statement:

Handling grocery shopping is extremely difficult in today's modern era because most of the people are too busy to go and buy food for themselves and family but in the case of non literate people or people with no knowledge of today's technology have difficulty to do grocery shopping even today when technology is at the fore front of innovation. The main reason for this issue is that most of the e-commerce and grocery shopping websites rely heavily on scripted assistance and text driven support. Hence people with little knowledge of technology or people with no knowledge of how to use such apps end up facing multiple barriers. Voice grocery is here to lessen and ease these problems stated above. Because this app supports voice interface which eradicates navigation through textual commands with in the application. In this app, users can search the desired item and add them in to the cart through simple voice commands in either English or Urdu language because many non-literate people in Pakistan are unable to speak or understand English Language. Hence this user-friendly design of application makes it more accessible to the users and makes this whole shopping experience more manageable for the non-educated and elderly users in world especially in Pakistan. It is a voice assisted user friendly solution that helps non literate individuals to do their own grocery shopping. The application also helps the users by spoken words recognition and feedback system which makes the purchasing experience easier to manage, thus mostly increase inclusion of a particular device in everyday life. We have taken a big step forward in narrowing the accessibility gap and creating a more equitable digital experience for everyone. This User-friendly voice-activated interface will help those non literate elderly people to do their own grocery shopping on daily basis without any accessibility gap. More over this app also supports the feedback system through simple spoken words in either English or Urdu language which makes this whole shopping experience easier to manage which will automatically 3 enhance participation of the individual in daily activities

independently. By choosing VoiceGrocery, we have taken substantial improvement in bridging this gap and allowing every individual to have an amazing and equitable grocery shopping experience.

1.2 Motivation:

The lack of competence and inability to shop for the grocery online by themselves causes autonomy and restricts their freedom as well as erodes their independence among the non literate and elderly folks in the society. VoiceGrocery is just a creative and groundbreaking method to overcome this disparity driven by the underlying vision and mission. VoiceGrocery aims to address this shortfall of receptiveness and accessibility by providing a voice based intuitive app. This app will help the folks to their grocery shopping without any help and guidance of any third party. As explained earlier it supports voice-based interface and virtual assistant, which will diminish the asking help of any other person. This technology will help consumers to do their daily grocery shopping without having the know how of how to read or write.

1.3 Problem Solution:

VoiceGrocery is committed to increase individual independence and self-sufficiency by providing a user friendly and very easy to use voice enabled way to achieve self-reliance and manage daily grocery needs. This app will be accessible on both android and iOS devices. Since it supports all the features of a typical e-commerce grocery app by simple voice commands, all hands free, making things easier for non literate and individual folks while online grocery shopping. Since non literate consumers often need someone to help them to guide while doing online grocery shopping, their ability to be self-independent and self-reliable is hindered, leading to a constraint liberty. Hence Voice grocery provides a solution which promotes self-confidence along with creating more digital space for individuals. This app also represents Groundbreaking step in advancing cross-accessibility solutions and provide an important tool to facilitate greater independence for those with limited literacy and less knowledge of technology.

1.4 Stake Holders:

The stake holders are as follows:

1) Primary Users:

The VoiceGrocery app's primary users are illiterate people and elderly folks, because such people have difficulty while using e-commerce grocery shopping apps because they support text-based assistance.

2) Development Team:

VoiceGrocery app is designed and maintained by the development team. This team may include front end developers as well as backend developers who will design the whole interface of the app and will maintain the functionality of voice-based commands and virtual assistant to improve the functionality of the app.

3) Supervisors and Project sponsors:

Supervisors who are responsible for providing guidance and ensuring the completion of the project within the given span of time are also included in the stakeholders. In context of VoiceGrocery, Ma'am Mariam Hida is the supervisor. Moreover, sponsors who are willing to provide financial or intellectual capital are also counted as stakeholders.

4) Retailers and Businesses:

Retailers can be stakeholders if they collaborate, by providing grocery products and businesses which will help coordinate the pickup and delivery function.

5) End Users and General Public:

People who prefer voice-based assistance while online grocery shopping, are counted as end users/stakeholders of this app.

6) Investors or Financial stakeholders:

In the future, if the idea becomes a commercial app, the involvement of investors will be very important, since these stakeholders will anticipate benefiting from their investment via commercialisation.

1.5 Project Description:

VoiceGrocery is a mobile app which is designed to bridge the convenience and exclusiveness for non-literate and elderly people. This app supports voice-based navigation and virtual assistance which helps perform the detailed shopping process like browsing the specific desired product, adding it to their cart, pickup and delivery process and feedback through simple voice commands—all hands-free. It is compatible on both Android and iOS devices which increases its comprehensive support. It also supports the use of advanced speech recognition in either Urdu or English language to provide effective and accurate answers. In this way, this app allows the non-literate and elderly folks to be independent in today's digital world.

1.5 Scope:

The VoiceGrocery aims to create a mobile app which is recognize voice speech through advanced voice speech recognition to make the shopping experience easy and accessible for non literate people. The target audience of this app includes non literate people, elderly folks and those who prefer voice assisted shopping. More over customers will have a virtual assistant which will help them while shopping like managing their shopping list, make payments etc. The goal is very important. VoiceGrocery will offer a hands-free shopping experience designed to empower non literate folks. Voice based navigation, virtual assistant and speech recognition are important features of these apps. Although apps like Daraz, dMart and food panda do exist in Pakistan and they excel in user friendliness and their functionality but these apps lack this crucial feature of voice-based navigation which will be supported by VoiceGrocery. Because this app is designed with accessibility in mind, hence even most illiterate people and those having zero knowledge of technology will be able to use it and order their desired food and products.

1.6 Modules:

The modules of the app are as follows:

Voice Assistant Module:

This is the core module of the application, where users talk to the app and it responds through voice commands.

1) Training the Voice Assistant:

The foundation of the Voice Assistant Module relies on accurate and comprehensive data. To make voice assistant work well, we need to train it with lots of Urdu phrases: including formal language and casual words etc. It will also learn to understand Urdu mixed with some English words (that are used in Urdu language as well). For example, when the user mention product names in English, the model will help the assistant to recognize different ways people speak as well.

2) Speech-to-Text (ASR):

In this module, the app will listen to what the user says and changes it into text. For example, if someone says, “mujhe chawal dekhao”, the app will turn

that into text so that it can understand the requests. We will make sure that the app understands different ways by which people pronounce words in Urdu.

3) Intents model (Understanding the user):

Once the speech is converted to text, it figures out what the user wants to do. Explaining the previous example, if someone says, “mujhe chawal dekhao”, the app will know that they want to search for rice. If they say, “isme sabsy acha konsa wala ha?” the app will understand that they want a comparison of different brands and qualities of rice. This model is trained to recognize different commands and requests.

4) Text-to-speech (TTS):

After processing the user's request, the app responds by speaking. TTS converts the textual response generated by the app into Urdu speech. For example, the app might say, "Yeh chawal aap ke liye behtreen hai" (This is the best rice for you). This makes the interaction smooth and easy for the user.

Client Module:

This is the user-facing interface where customers can interact with the app aside from the voice features.

1) User Interface and User Experience (UI/UX):

While the app is voice-centric and most things are voice-controlled, it still provides a simple user interface for basic functions like registration, profile management, and settings. For people who cannot read (illiterate) or elderly users, the design will be super minimal with large buttons and easy-to-understand visual aids (e.g., product images with minimal text). Our goal is to make sure that anyone can use the app easily, even if they are not used to technology.

2) Shopping Cart & Checkout:

The app will automatically add the products and items to the cart when the user tells it to. After they are done shopping and ready to buy, the app will

guide them through the checkout process using voice commands. It will also confirm the total cost and asking about the delivery and payment choices.

Delivery Module:

This part handles everything about delivering groceries to our respected customers.

1) Order Assignment:

Once the user has placed the order, the app will automatically assign it to the closest delivery person, making sure the order gets to the customer quickly.

2) Order Tracking:

This module allows the user to track their order through voice commands like "Mera order kahan tak pohuncha hai?" (Where has my order reached?). The system will respond with live updates on the delivery status, giving the user a sense of control over the delivery process.

3) Delivery Confirmation:

When the groceries are delivered, the app will ask the user to confirm that they received everything. It will help making sure all deliveries are completed correctly.

Admin Module:

The Admin Module provides the necessary tools for app administrators to manage the platform.

1) Product Management:

Administrators will have the ability to add, edit, or remove products from the catalog, ensuring that the latest items are available for users. This includes adding product details, pricing, availability, and descriptions.

2) Order Management:

Admins can keep an eye on all ongoing and past orders, resolve issues, and handle customer queries. If a problem arises with an order, such as an unavailable product, admins can contact the user through automated voice notifications or directly intervene to resolve the issue.

3) User Management:

Admins can also manage user accounts, including the registration and

preferences. They can check how users are using the app to make things better.

1.7 Tools and Technologies:

1) Front end development:

Flutter: Flutter will allow us to build app which will be accessible on both android and iOS.

Figma: Figma will be used to design the UI/UX part of our app to help make app visually better to make it under stable for non literate users.

React: react will also be used to design the UI/UX part of our app.

2) Back-end development:

Firebase: will be used for data management of our app. And it will also provide user Access control and service review in VoiceGrocery.

SQLite: All the database of our app will be supported by SQLite featuring offline access to shopping list and other important features of VoiceGrocery.

3) Real time Integration:

Flask: Real time integration will be supported by APIs in Flask to manage and control interaction between users and data flow in the VoiceGrocery app.

Python: will be used for the backend logic. Like it will support the correct voice speech recognition and virtual assistant features of VoiceGrocery.

4) Machine learning:

TensorFlow: Will be used for the development for models which will recognize the voice commands it will also support the virtual assistant feature in VoiceGrocery app.

PyTorch: will be used to train and fine-tune models. In our app it will be used to understand the complex voice commands and process them afterwards.

5) Management tools:

Trello: Trello will be used to organize our weekly tasks and iterations ensuring a well-coordinated app and timely development of that app.

6) Version Control:

GitHub: will be used to upload the source code ensuring teamwork and optimal

alterations during the whole development process of VoiceGrocery.

1.8 Work Division:

Modules	Hasan Kamal	Fatima Mazhar	Naima Zafar
Data Acquisition	✓	✓	✓
UI/UX		✓	✓
Delivery		✓	
Checkout			✓
Database Creation	✓		
Search and browsing			✓
Reviews		✓	
Model Training	✓		
Model Integration	✓		

1.9 Timeline:

Phase-1: September-October

During these two months, the main focus of our app is building the UI/UX of the VoiceGrocery app, including the prototype of the UI interface to make sure that it is user friendly and completely understandable to non literate users. More over data collection is also part of this phase. Like whole data which will be used afterwards will be collected and prepared to train the model. This time span/phase is the basis for the usability of VoiceGrocery.

Phase-2: November-December

The 2nd phase is marked to train the model on the data we collected in the phase 1. The main model will be trained and developed in this 2 month's time span. Various machine learning models which process the user commands (easy and difficult) will be trained to understand in this phase. Meanwhile the application database will also be created to store important

information of the user. Along with some important features such as payment and pickup/delivery process will also be integrated which forms the basis of VoiceGrocery.

Phase-3: January-March

The third phase is dedicated to improving the app from the initial user feedback from the last 2 phases. And then improving it afterwards. More over in this time phase some more important features will be integrated as well such as cart management and delivery/pickup options, and allowing users to complete orders with voice commands. Mainly the focus will be on optimizing the app for use and preparing it for full release.

Phase-4: April-May

In the final phase, the app will be evaluated and refined ed for the final release. This phase will include thorough testing and comprehensive evaluation to identify bugs or issues considering the optimization as well to ensure that the app works well and is user friendly and completely accessible to users on different devices. After the app is refined and validated, the final version will be released.

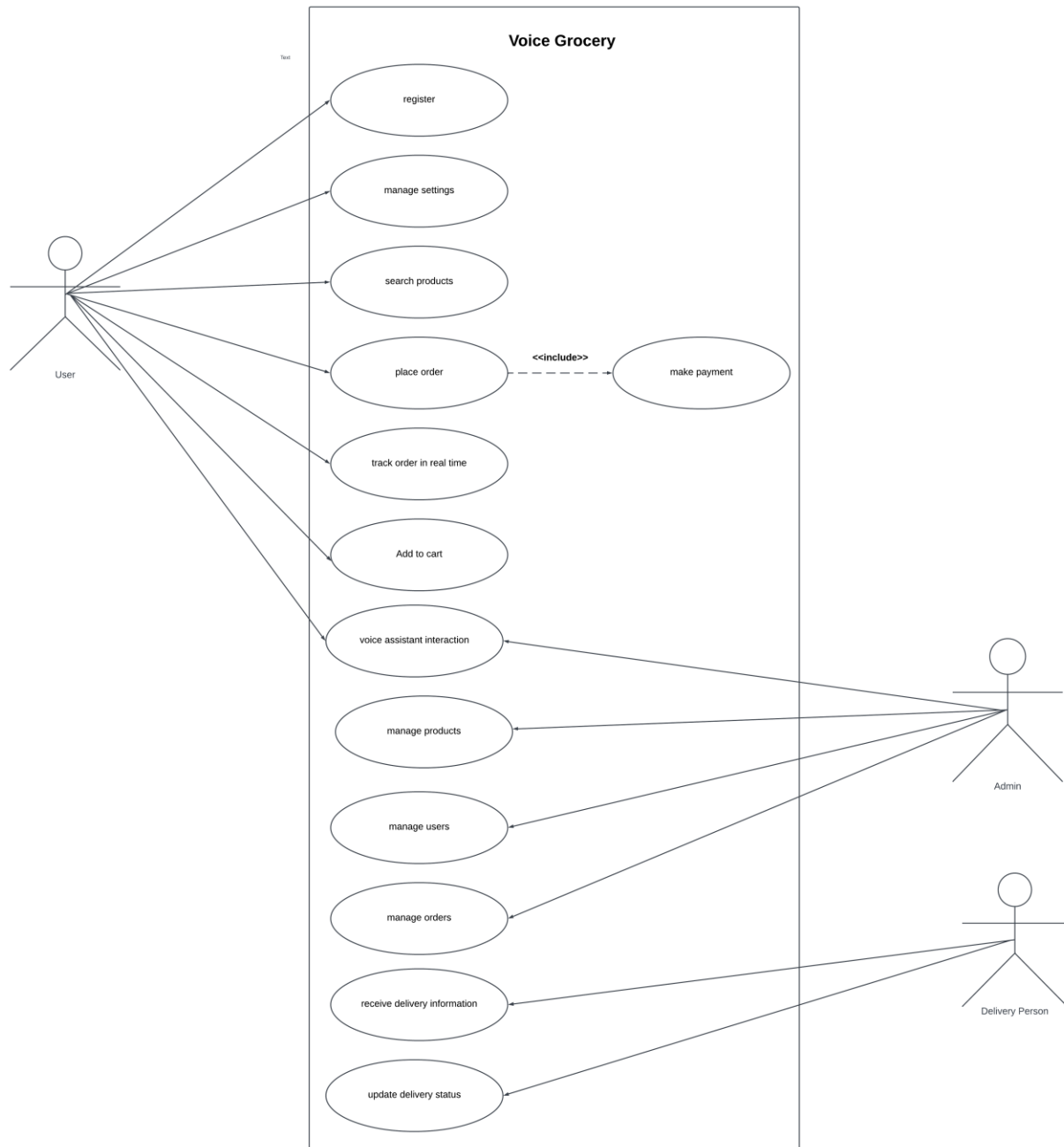
2 Project Requirements:

In today's modern digital world, where everything is altered and almost every product is just a click away, some illiterate people still find grocery shopping difficult. Because these grocery apps rely heavily on text-based guidance. Which increases the difficulty for such people when it comes to shopping for food online. **Voice Grocery** solves this major issue by offering simple voice-based navigation and a virtual assistant where the user can control the whole process through simple voice commands, and the user can search for the specific process, add it to the cart, then check out; moreover, the user can also give audio feedback about the product he/she has received. This strategy bridges the gap in engagement of users in daily routine tasks such as food and promotes independence and autonomy among the users.

2.1 Use Case Diagram:

The key functionalities of our app include user authentication, voice-based navigation, voice assistant interaction, product search and browsing, speech-to-text (ASR) done, text-to-speech (TTS), shopping cart management, payment and delivery, and order tracking, which will be handled by the user. Whereas, user profile management, managing products and orders, resolving customer queries, and all the sales analytics will be looked at by the admin. In terms of non-functional requirements, the **Voice Grocery** app is built to be highly usable, and it will perform well in all terms, providing scalability, security, compatibility, reliability, localization, user

experience consistency, and data privacy to all its users. In other words, the **use-case** for our app can be viewed as:



2.2 Functional Requirements:

Let's start with the functional and non-functional requirements for each module:

Voice Assistant Module:

Functional Requirements:

- 1) The system shall recognize and process voice commands in both Urdu and English.
- 2) The system shall convert speech to text accurately.
- 3) The system shall interpret user intents from the converted text.
- 4) The system shall generate appropriate responses based on user intents.
- 5) The system shall convert text responses to speech in Urdu.

Non-Functional Requirements:

- 1) The speech recognition system shall have an accuracy rate of at least 95% in normal speaking conditions.
- 2) The system shall process voice commands and generate responses within 2 seconds.
- 3) The system shall support a variety of Urdu accents and dialects.

Client Module:

Functional Requirements:

- 1) The system shall provide a simple, visual user interface for basic functions like registration and settings.
- 2) The system shall allow users to manage their shopping cart using voice commands.
- 3) The system shall guide users through the checkout process using voice instructions.
- 4) The system shall confirm order details and total cost verbally before finalizing an order.

Non-Functional Requirements:

- 1) The user interface shall be accessible to users with low literacy levels.
- 2) The system shall support both Android and iOS devices.
- 3) The app shall be usable by 90% of test users without prior training.

Delivery Module:

Functional Requirements:

- 1) The system shall automatically assign orders to the nearest available delivery person.
- 2) The system shall provide real-time order tracking information upon user request.
- 3) The system shall prompt users for delivery confirmation upon order completion.

Non-Functional Requirements:

- 1) The system shall update order status in real-time, with a maximum delay of 30 seconds.
- 2) The order assignment algorithm shall optimize for the shortest delivery time.

Admin Module:

Functional Requirements:

- 1) The system shall allow administrators to add, edit, and remove products from the catalog.
- 2) The system shall provide tools for managing and monitoring orders.
- 3) The system shall allow administrators to manage user accounts and analyse usage patterns.

Non-Functional Requirements:

- 1) The admin interface shall be secure, requiring multi-factor authentication for access.
- 2) The system shall log all administrative actions for auditing purposes.

2.3 Non-Functional Requirements:

Some other Non-Functional requirements can be:

Availability:

- 1) The system should have an uptime of 99.9%, ensuring high availability for users to shop at any time without downtime.
- 2) Backend services should be robust with failover mechanisms to ensure seamless transitions in case of server issues.

Error Handling:

- 1) The app should gracefully handle errors and provide clear voice feedback. For instance, if a voice command is misunderstood, the system should prompt the user to try again or clarify the command.
- 2) Connectivity issues or temporary service failures (e.g., payment gateway issues) should notify the user with a clear error message or voice prompt while maintaining their progress in the app.

Data Integrity:

- 1) Transactions, order data, and user profiles should be consistently backed up to ensure that no data is lost in the event of a system crash or failure.
- 2) The app must prevent duplicate orders or cart items from being added due to unexpected interruptions.

Usability:

- 1) The app shall be usable by individuals with no prior experience with smartphone applications.
- 2) 90% of users should be able to complete a purchase within 5 minutes of their first use.
- 3) The app shall provide clear voice feedback for all user actions.
- 4) The user interface shall use large, easily recognizable icons to supplement voice interactions.

Security:

- 1) All data transmissions shall be encrypted using industry-standard protocols (e.g., TLS).
- 2) User passwords shall be hashed and salted before storage.
- 3) Secure login methods like **OAuth** or multi-factor authentication should be in place for user accounts to prevent unauthorized access.
- 4) The app shall implement secure authentication mechanisms.
- 5) Regular security audits and penetration testing shall be conducted.
- 6) The app shall automatically log out inactive sessions after 15 minutes.
- 7) User data such as profile information, order history, and payment details must be encrypted, and strict privacy policies must be enforced.

Performance:

- 1) The app should process and respond to voice commands within 1-2 seconds for common tasks like product search, adding items to the cart, or providing feedback.
- 2) Speech recognition should process voice inputs with minimal delay, especially for users with different accents or languages.
- 3) The system should scale efficiently to handle multiple users interacting with the voice assistant at the same time, without affecting response time.
- 4) Should handle a growing number of products and orders without significant performance degradation.
- 5) The backend services should maintain consistent performance under high traffic, especially during peak shopping hours or sales events.
- 6) Load balancing techniques can be employed to distribute traffic evenly across servers.

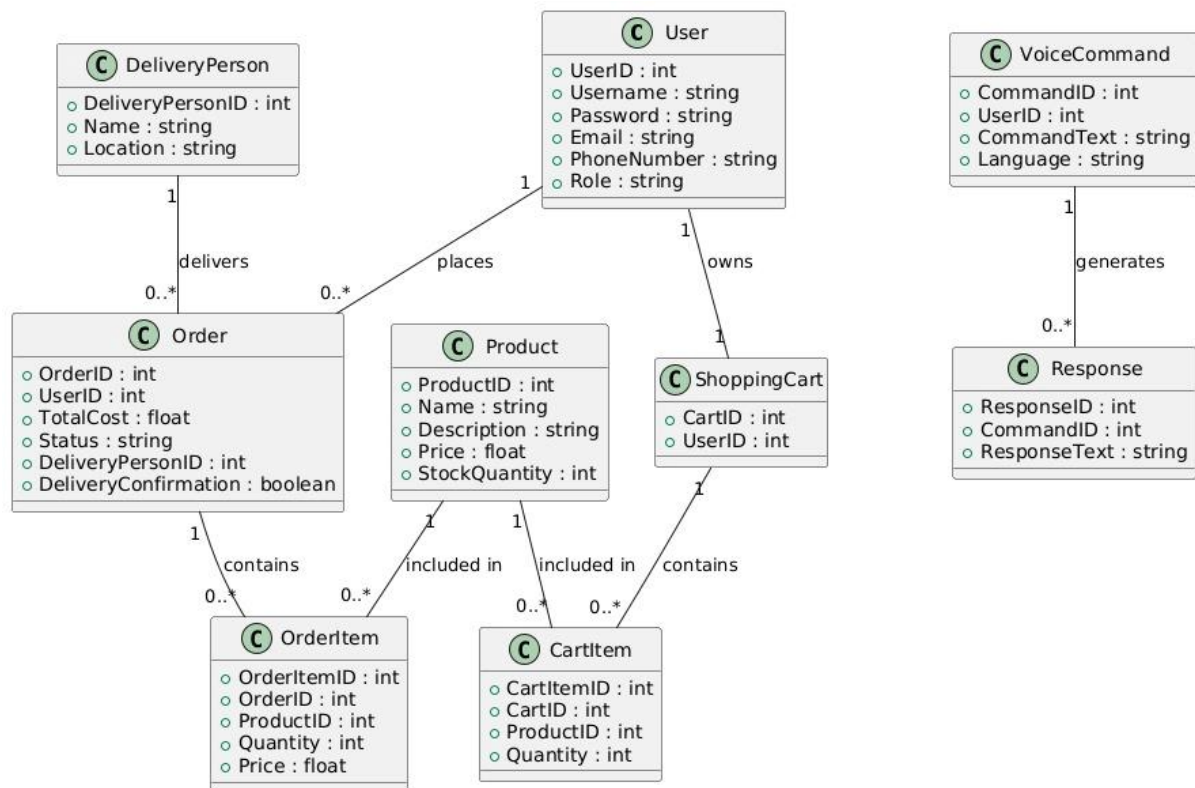
Maintainability:

- 1) The app should follow a modular design, where different components (e.g., voice assistant, cart management, payment processing) can be updated independently to facilitate easier maintenance and updates.
- 2) The system architecture should be flexible enough to allow new features, such as additional languages or voice APIs, to be added without significant rework.

Testability:

- 1) The system should be designed to support unit and integration testing of key components, such as voice processing, product search, and payment.
- 2) Voice interactions should be tested across different accents and languages to ensure high accuracy and natural flow.

2.4 Domain Model:



The domain model explains the classes which will be added in our app along with the relationships between those classes.

3 System Overview:

VoiceGrocery is a mobile application designed to empower visually impaired users and those with limited literacy by providing an accessible, user-friendly, and voice-operated grocery shopping experience. The app leverages **voice recognition technology** and a **virtual assistant** to facilitate the entire grocery shopping process, from browsing products to making purchases, tracking orders, and managing deliveries.

The app's primary goal is to create a barrier-free shopping experience, allowing users to interact with the system through **natural language commands** in both **English and Urdu**, without the need for traditional visual interfaces or text-based inputs. This significantly enhances autonomy and ease of use for individuals who may struggle with conventional shopping apps due to visual or literacy constraints.

3.1 System and Architecture Design:

The **VoiceGrocery** system comprises multiple layers that work together to provide seamless functionality:

- **Frontend (Mobile App):**

The app interface is designed using cross-platform frameworks like **React Native** or **Flutter**, ensuring compatibility with both Android and iOS devices. Users interact with the system using **voice commands** facilitated by an intuitive interface that requires minimal visual engagement.

- **Voice Processing Layer:**

The app integrates with advanced **Speech-to-Text (ASR)** and **Text-to-Speech (TTS)** engines, converting spoken commands into actionable inputs and providing feedback in natural language. **Natural Language Processing (NLP)** is employed to interpret user commands, extract intent, and facilitate interactions in both English and Urdu.

- **Backend (Business Logic):**

The core backend handles **user management, product management, order processing, payment integration, and delivery scheduling**. This layer interacts with external services like payment gateways and delivery platforms to ensure a smooth workflow.

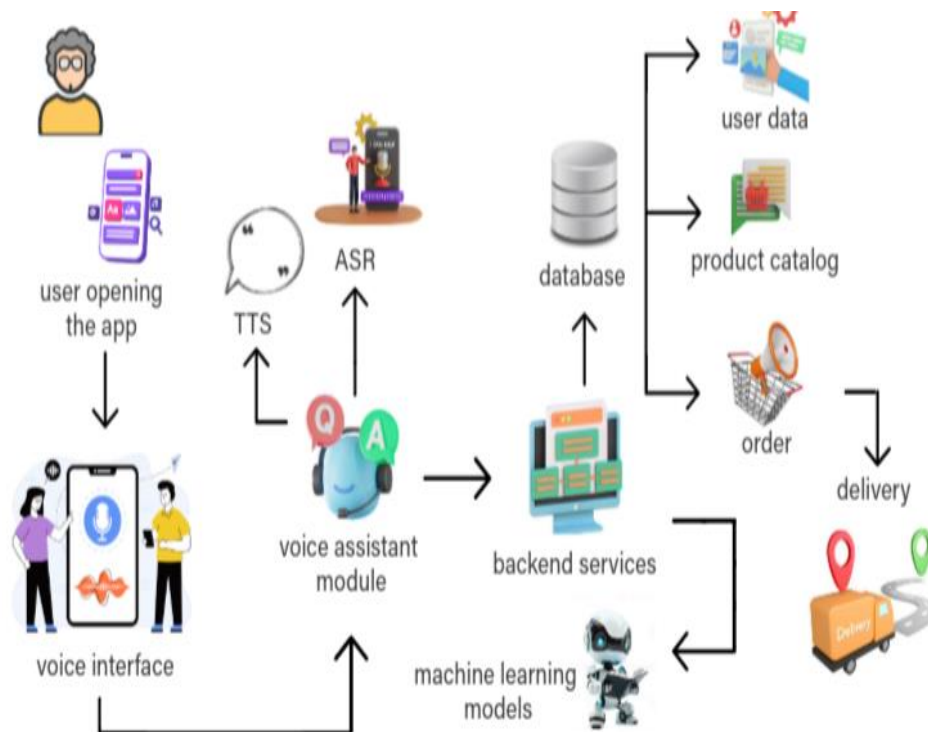
- **Database:**

A centralized **database** stores user profiles, product catalog, order histories, and preferences. This data is accessed in real-time to personalize the shopping experience for individual users.

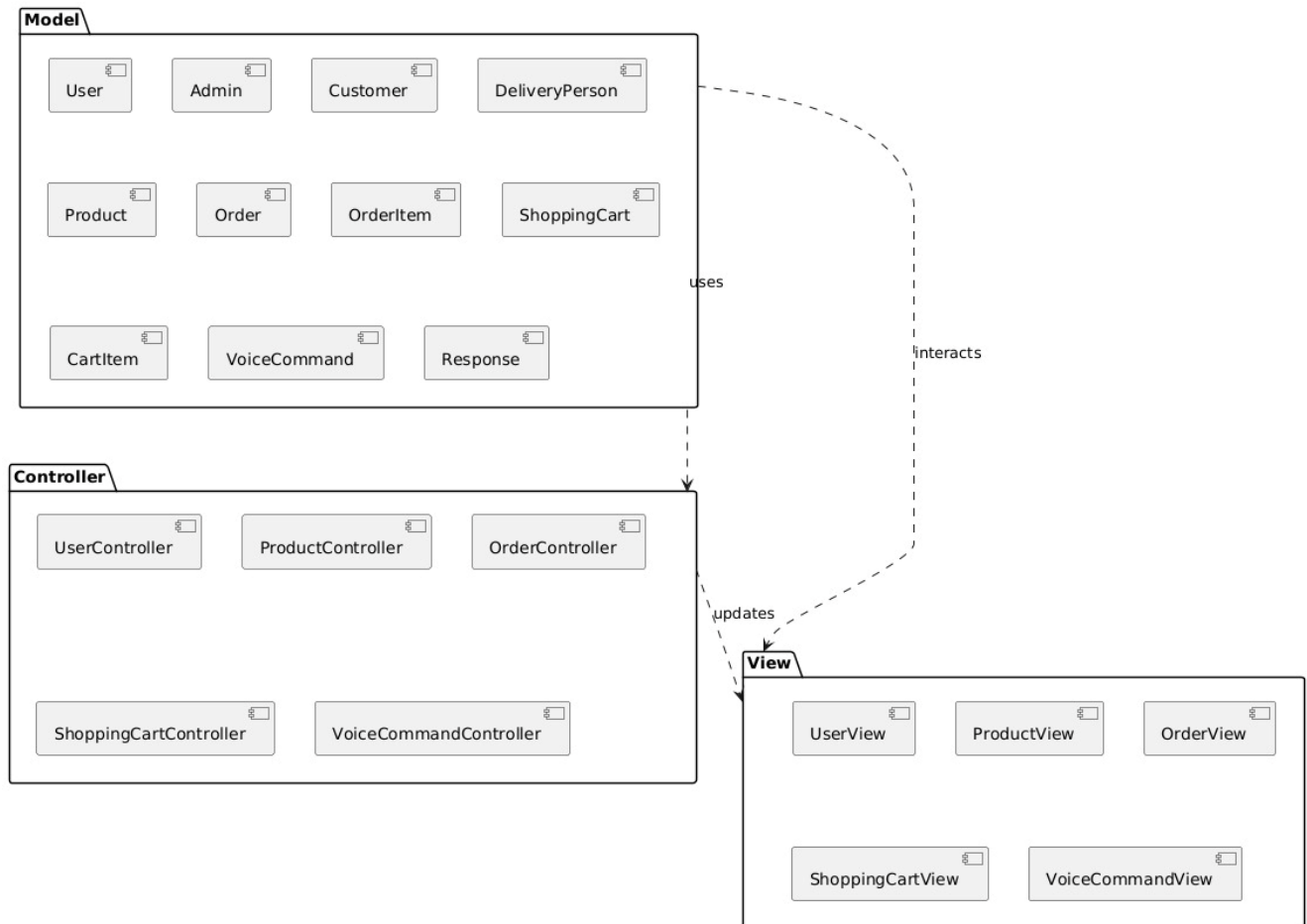
- **External Integrations:**

VoiceGrocery integrates with third-party APIs such as **payment gateways** (e.g., Stripe, PayPal), **delivery services**, and voice-processing platforms (e.g., Google or AWS services) to provide voice recognition and payment functionality.

Architectural Design:



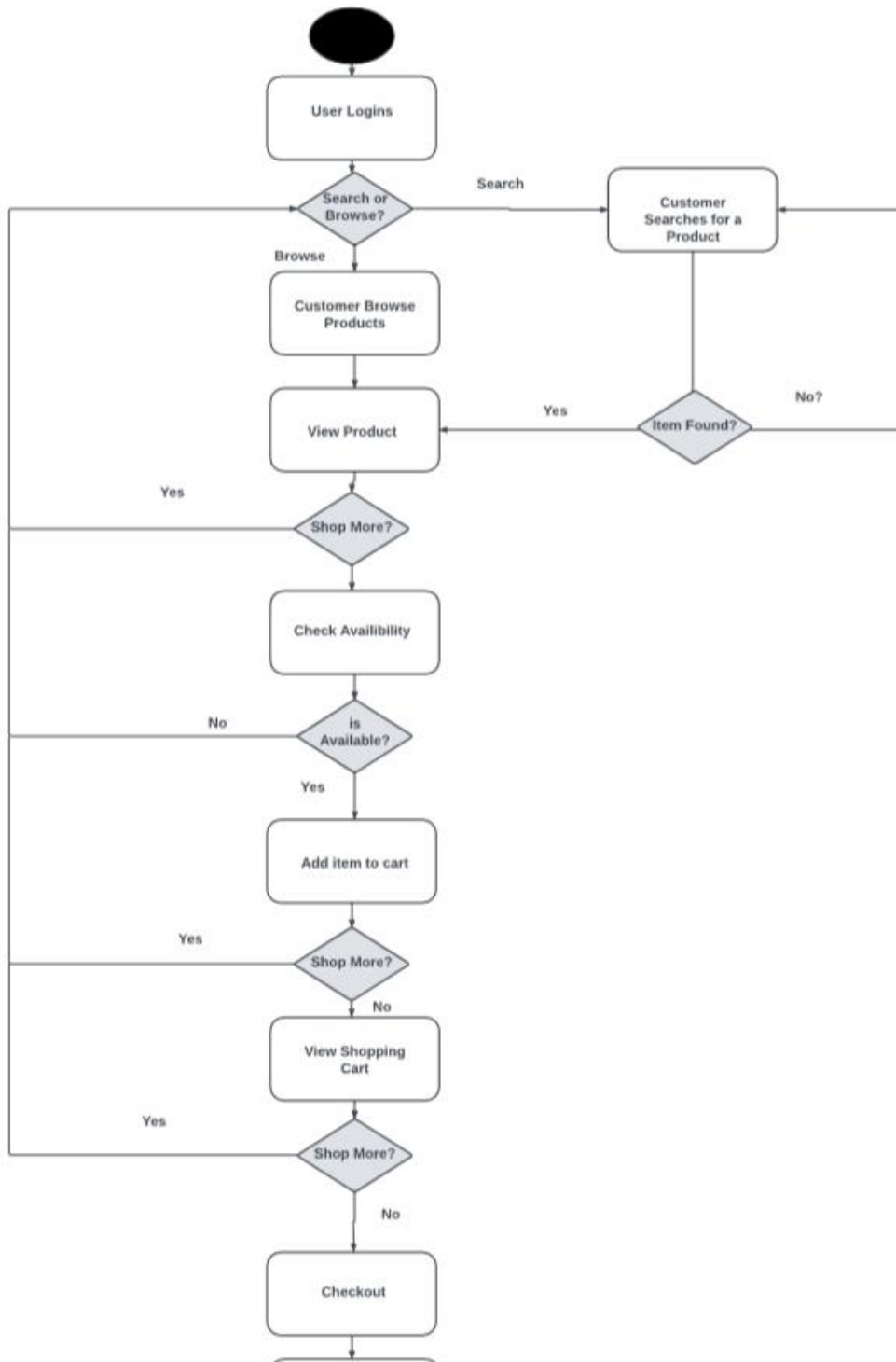
Package Diagram:

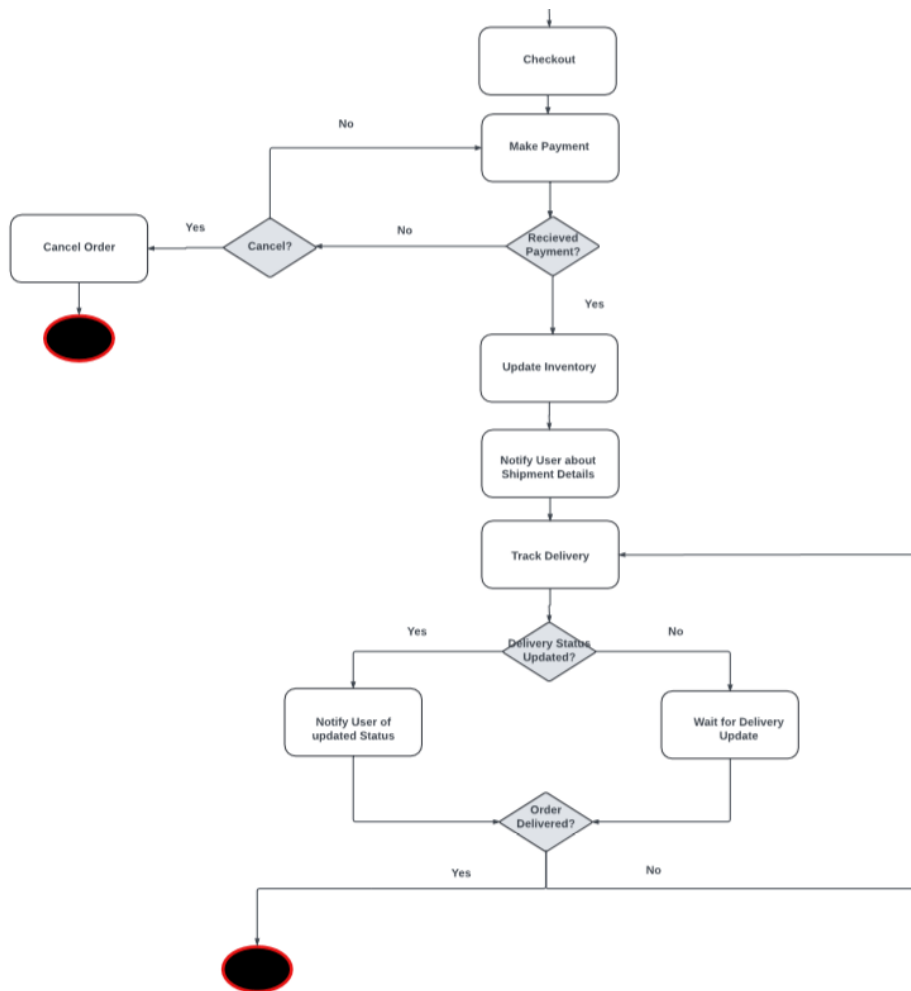


The package diagram explains the software design pattern, as our app is user based so we have used MVC design pattern through which user can easily use our app having these three distinct elements known as model, view and controller as shown above.

3.2 Design Models:

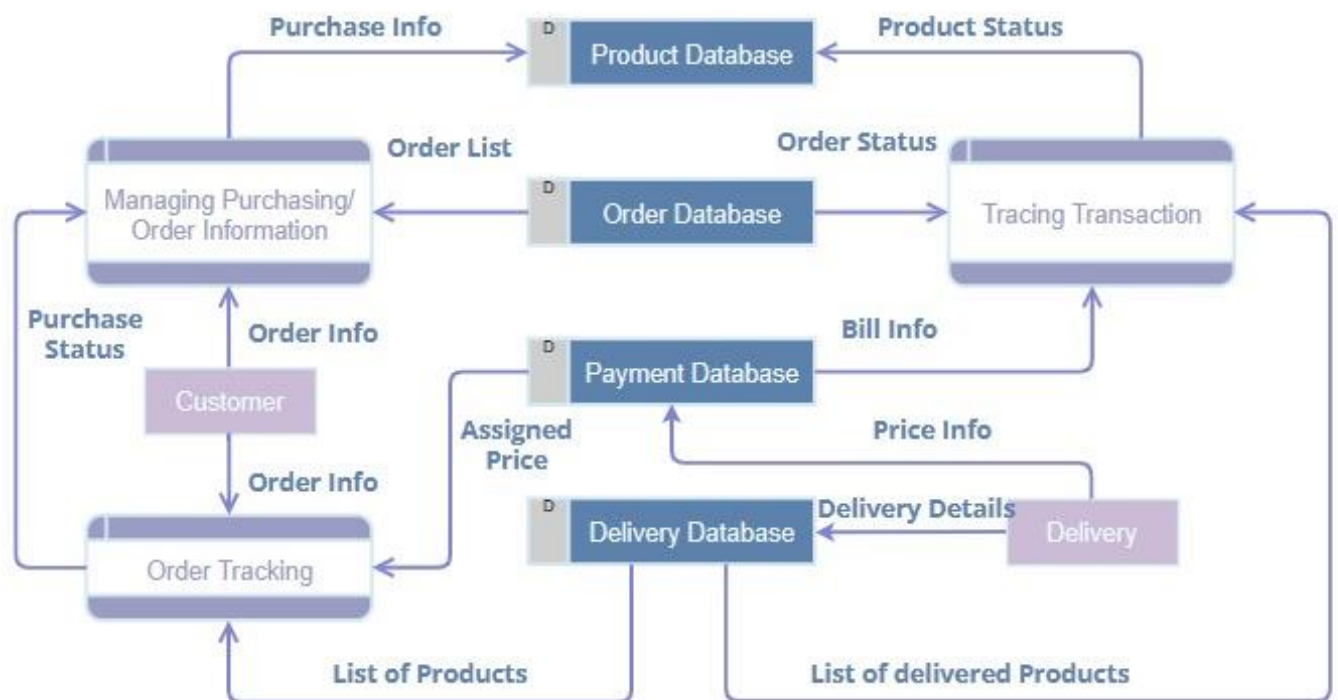
Activity Diagram:



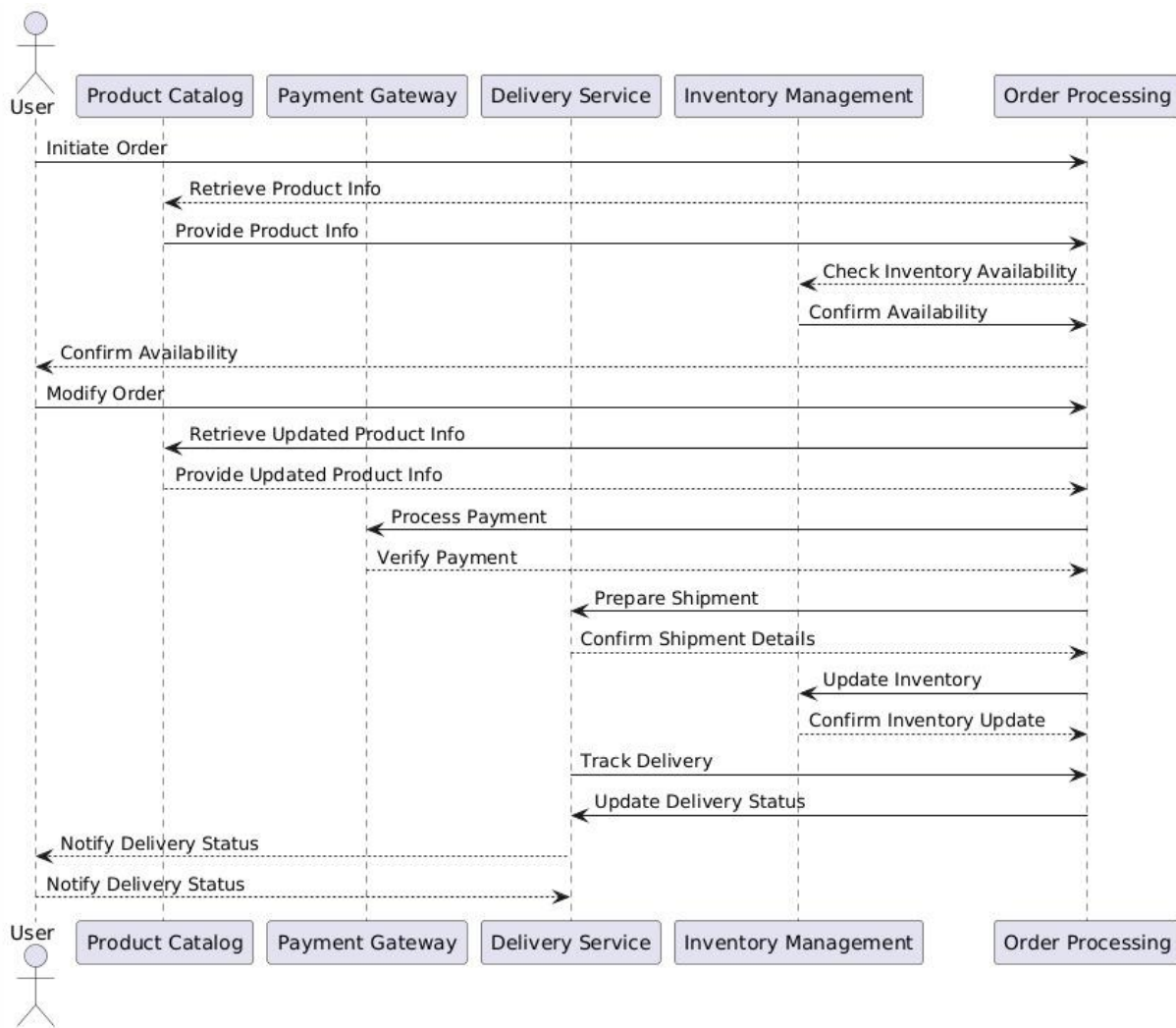


The activity diagram explains the order in which the whole process of our app is explained in a flow.

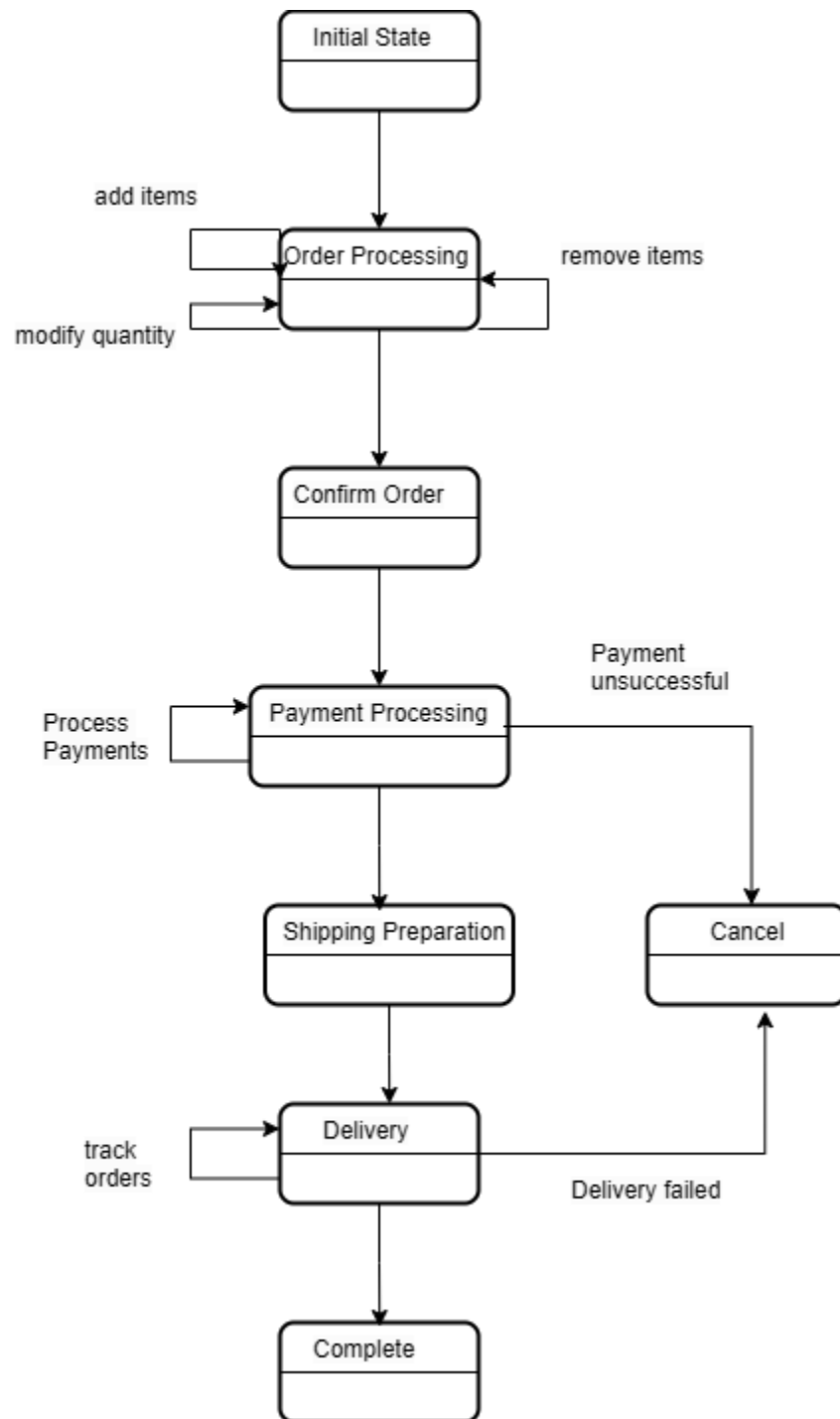
Data Flow Diagram:



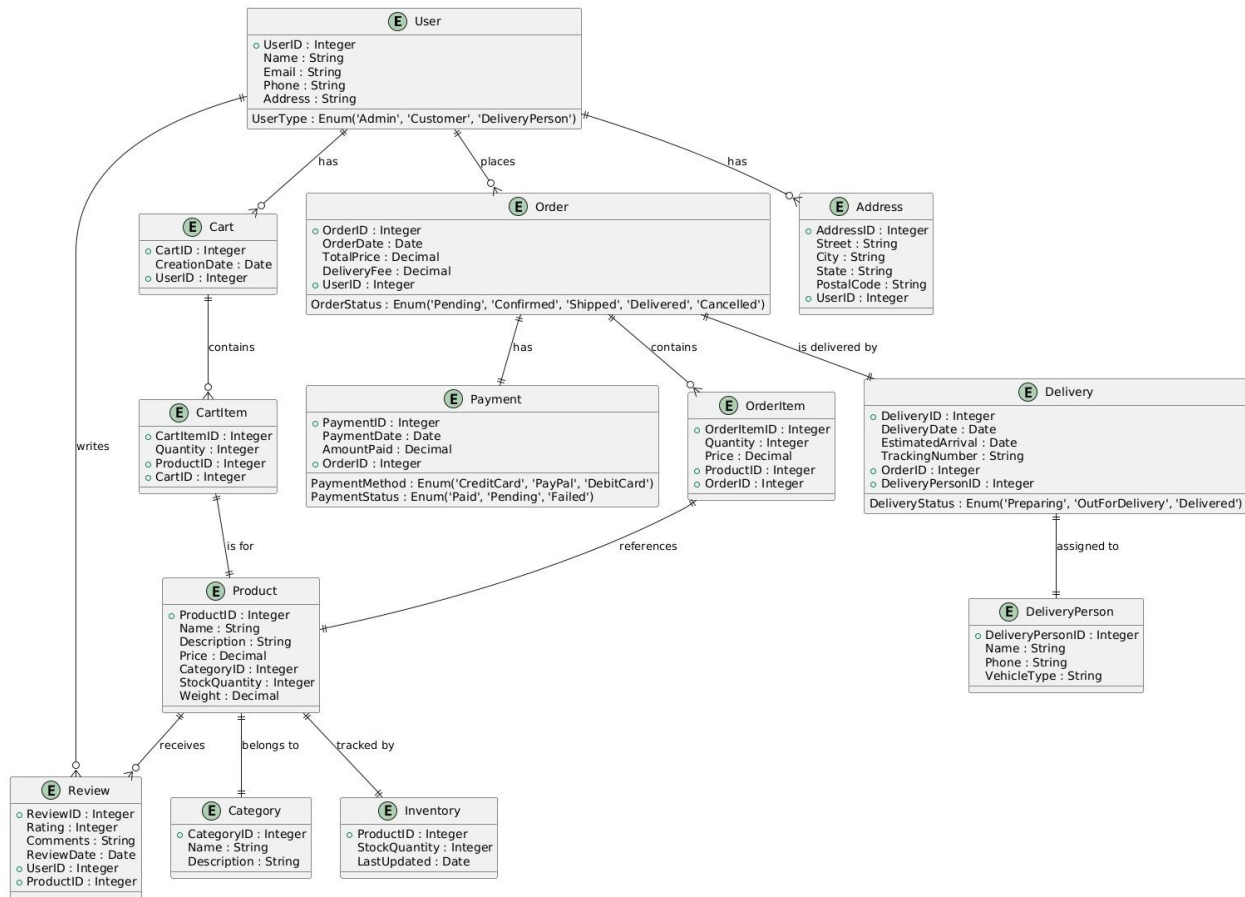
System-level Sequence Diagram:



State Transition Diagram:



3.3 Data Design:



This diagram outlines the data design for your **VoiceGrocery app**, showing the relationships between key entities:

- **User**: Stores user details and types (Admin, Customer, DeliveryPerson).
- **Cart**: Linked to users, contains products through **CartItems**.
- **Order**: Represents purchases made by users, including **OrderItems**, delivery, and payment information.
- **Product**: Contains product details, belongs to a **Category**, and is tracked in **Inventory**. Products can also have **Reviews**.
- **Delivery**: Manages the delivery of orders by assigned **DeliveryPersons**.
- **Payment**: Tracks payment methods and statuses for orders.

This structure supports the app’s functionalities like placing orders, managing carts, and handling deliveries.

4 Conclusion:

The **Voice Grocery app** successfully addresses the accessibility challenges faced by non-literate individuals in grocery shopping. With features like **voice-based navigation**, **speech recognition (STT)**, and **text-to-speech (TTS)** in both English and Urdu, it enables users to shop independently and interact with the app hands-free. The app ensures an intuitive, user-friendly experience, helping non-literate people access essential services with ease.

Future work can focus on expanding the app's language support, enhancing personalization through machine learning, integrating with more local stores for real-time updates, and improving security with voice biometrics. These developments will further enhance accessibility and user satisfaction.

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